

Digitally Secure

If all those paintings by the masters and the antique Wedgewood and Noritaki adorning your house leave you feeling insecure during long absences from home, broadband can come to your rescue. You can use a security network that consists of a few inexpensive Web cams that can stream live images for you to view in a browser from anywhere across the world. And this is no remote possibility. It's happening right now. Vicinium offers a NeighborMation server which lets you arm, disarm, and monitor your security system status remotely. It can also send security alerts to trusted neighbours in case of suspicious activity and automatically arm their security systems. Pretty neat, huh?

now a considerable amount of interest from some of the major computer and networking manufacturers.

HomePNA is an association of several companies that include Compaq, AMD, AT&T, Intel, Motorola and HP that are working together to ensure adoption of a single, unified phone line networking standard. All this leads to the conclusion that in the not too distant days to come, you won't need to be Bill Gates with \$50 million to spend to have an intelligent digital home. Home gateways will form an important part of the home of the future. The current generation of home gateways allows multiple users to connect to the Internet, with numerous companies offering solutions that act as combination proxy servers and firewalls. However, the next generation of home gateways promises to deliver a lot more.

The Open Services Gateway Initiative (OSGi) is an independent consortium with more than 75 members from around the globe. This group is working to define open specifications for the delivery services over wide-area networks to local networks and devices. The OSGi framework will allow multiple software services to be run off a services gateway. This could be a set top box, cable/DSL modem, or even a dedicated residential gateway. Sun Microsystems has introduced an OSG-compliant Java Embedded Server that will allow people to access subscription-based services over a network. IBM is also developing home gateway technology, based on WebSphere embedded software, along with National Semiconductor Corp. This aims at extending their vision of pervasive computing which involves computerising anything from refrigerators to a pair of earrings.

No software, no go

For a truly intelligent home you could end up needing huge amounts of processing power and storage space in addition to expensive server software. Realistically speaking, it is unlikely that most ordinary households will have this. Intelligent networks are what will power these homes if they are to be accessible to a larger number of people. The multiplicity of the hardware and the software platforms that form networks make a cross-platform architecture absolutely necessary. This means that not only does software need to be able to run on different hardware and operating systems, but also services written for different platforms need to be able to communicate with each other. It becomes necessary that many of the features that are offered be provided by service providers. This would mean that the function of processing is actually distributed amongst various sections of the network. Java, from Sun, allows developers to



An AMX control system

write applications for a device without worrying about what kind of hardware and software is involved. This is very important because it cuts down the development time for any software application, which is the heart of any service provided over a network.

Jini is one network technology introduced by Sun that enables devices to plug together to form a community. This happens without prior planning and devices provide services to other devices that are a part of the community. New devices on the network register themselves with a lookup service, which in turn provides information that allows a user or another device to detect them. Microsoft's new .Net platform, which aims at enabling XML-based Web services is also likely to play an important part in the future.

There are a number of developments taking place in embedded operating systems that are worth taking note of. Linux players like RedHat are becoming increasingly active in this area. Numerous other players are also trying to get their technologies established.

Welcome home

The Jetsons' home may still sound like a futuristic fantasy to most of us, but it's closer than you think. While the reality is that there is still some way to go before the average home has the kind of intelligence that is on display in the cartoon series, any new home that you buy, five to 10 years from now, will include a host of connectivity and convenience options. Existing homes are already being fitted with smarter appliances and electronics.

The fact is that almost every prominent technology company—be it IBM, Intel, Sun, Seagate or Microsoft—is involved in projects that demonstrate the use of technology in homes. Consumer electronics majors and traditional home automation companies also seem to be trying their best to develop new products that will excite homeowners. So, are you wondering when exactly you are going to see all of this in your house? Well, the future is out there and it's waiting to happen. ■

<http://www.>

http://architecture.mit.edu/house_n/web/index.htm

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www.techhome.org

Integrated Home Systems

www.cisco.com/warp/public/3/uk/ihome/

Cisco: The Internet home